ET AI HACKATHON 2.0 · ROUND 2 SUBMISSION

Industrial Knowledge Intelligence

A Hybrid GraphRAG Platform for Heavy Industry

Problem Statement #8 — Unified Asset & Operations Brain

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1. Executive Summary

Industrial plants generate vast quantities of documents — engineering drawings, maintenance logs, inspection reports, safety procedures, incident records — and scatter them across separate systems that do not talk to each other. When a decision has to be made, the knowledge needed is often present in the organisation, just not reachable in time. This submission builds the intelligence layer that makes that scattered knowledge queryable.

What this system does

A field engineer asks a question in plain language and receives a cited, confidence-scored answer in seconds. The system reasons across documents that share no common keywords — reconstructing, for example, the full early-warning chain behind an equipment failure from five separate records that keyword search would never connect.

The prototype is a working Hybrid GraphRAG platform: it combines a knowledge graph for relationship reasoning with a vector store for semantic similarity, fuses the two result sets using Reciprocal Rank Fusion, and uses a large language model to synthesise a final answer with citations back to the source document. It runs end-to-end on a real document corpus, is fully reproducible, and is benchmarked with hand-labelled ground truth against a keyword-search baseline.

2. The Problem

A 2024 McKinsey survey found that professionals in asset-heavy industries spend an average of 35% of their working hours searching for information or recreating documents that already exist somewhere in the organisation. In India specifically, a NASSCOM-EY study found the average large plant runs across 7 to 12 disconnected document systems — drawings in one place, work orders in another, procedures in a third, inspection records in a fourth, regulatory submissions scattered across email.

This fragmentation contributes to an estimated 18 to 22% of unplanned downtime events in Indian heavy industry, because maintenance teams make decisions without complete equipment history. Compounding it is a knowledge cliff: an estimated 25% of India's experienced industrial engineers will retire within the next decade, taking decades of undocumented operational knowledge with them. Once that knowledge leaves the building, it cannot be recovered.

> **The core failure mode**
>
> The problem is rarely that data is absent. It is that data is present but unacted upon — the signals exist across separate records, but no intelligence layer connects them into a decision in time. This is the exact pattern behind well-documented industrial incidents, where warning signs were recorded but never assembled into a picture anyone could act on.

3. Solution Overview

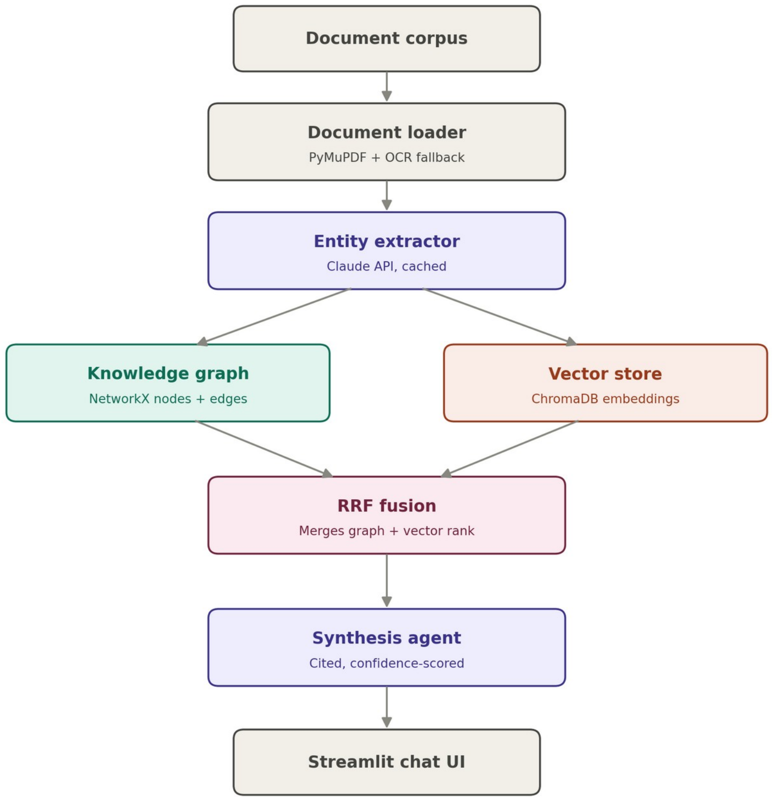
The system ingests heterogeneous industrial documents and turns them into a single queryable intelligence layer. Two capabilities are built end-to-end:

- **Universal document ingestion and knowledge graph.** The pipeline parses PDFs and scanned forms, extracts typed entities and the relationships between them, and assembles a knowledge graph spanning the whole corpus, deduplicating entities that appear across many documents.
- **Expert knowledge copilot.** A retrieval layer combines graph traversal with semantic search, and a synthesis agent produces a natural-language answer with a citation to the source document and a confidence rating, delivered through a mobile-friendly chat interface.

Three further components — a maintenance and root-cause agent, a regulatory compliance mapper, and a lessons-learned engine — are specified as a roadmap rather than built, keeping the prototype deep on two capabilities rather than shallow across five.

4. Architecture

The system is a seven-stage pipeline. A document enters at the top; a cited answer leaves at the bottom. The defining feature is the fork and merge in the middle: the knowledge graph and the vector store are built and queried independently, then their results are fused. This is what makes the system hybrid rather than a plain vector-RAG wrapper.



Hybrid GraphRAG pipeline — Industrial Knowledge Intelligence

Each stage is a standalone module, so any single layer can be swapped without disturbing the others — for example, replacing the in-memory graph with a production graph database, or the local vector store with a managed service, changes one module and nothing else.

Component stack

Stage	Technology	Purpose
Document loader	PyMuPDF, pdfplumber, pytesseract	Parses PDFs with layout and page boundaries preserved; falls back to OCR automatically for scanned pages.
Entity extractor	Claude API (claude-sonnet-4-5)	Reads each document and extracts typed entities and their relationships as structured JSON. Cached

Stage	Technology	Purpose
		to disk so re-ingestion is free.
Knowledge graph	NetworkX	Stores entities as nodes and relationships as edges. Merges entities by canonical ID so one asset referenced many ways resolves to a single node.
Vector store	ChromaDB, sentence-transformers	Embeds document text locally for semantic similarity search. Catches answers the graph missed. No per-chunk API cost.
Hybrid retriever	Reciprocal Rank Fusion	Queries graph and vector store in parallel and fuses the two ranked lists by rank score.
Synthesis agent	Claude API (claude-sonnet-4-5)	Turns fused retrieval results into a final answer, attaches a source citation, and scores its own confidence.
User interface	Streamlit	Chat interface that renders on mobile and shows the answer, citation, and confidence score inline.

5. Why Hybrid, And Why Not The Alternatives

Why not pure vector RAG

Vector RAG is what most teams ship. It handles semantic lookups well but is blind to relationship reasoning. Asked which procedures relate to an incident and which regulation covers them, it returns chunks that look similar to the query rather than reasoning over actual relationships. Multi-hop questions — the ones that matter most in industrial knowledge work — are exactly where it fails.

Why not pure graph RAG

A vectorless graph-only system is elegant but brittle in a short build. If entity extraction misses something during ingestion, that information is gone with no fallback. Keeping a vector store as a safety net means the system can still recover an answer from raw text even when the graph did not capture it.

Why hybrid

Recent enterprise research documents that hybrid GraphRAG — fusing vector similarity with graph traversal via Reciprocal Rank Fusion — outperforms both standalone vector RAG and standalone graph RAG on retrieval accuracy and answer quality. This is the empirically stronger architecture, and it maps directly onto two of the named judging criteria: knowledge graph linkage completeness, and query answer quality.

6. Design Lineage

This architecture was not learned for the hackathon. It re-points a pattern the author already operates in production. A custom agentic system built for a live algorithmic-trading product uses the same three-layer shape — a memory layer over a knowledge corpus, a prediction and scoring layer, and an orchestration agent on top. Each maps directly onto a component here.

Production system (trading)	Role there	Equivalent here
Persistent memory layer	Holds and connects the knowledge corpus	Knowledge graph (NetworkX) plus the extraction cache
Prediction and scoring layer	Ranks and scores candidate outputs	RRF fusion plus answer confidence scoring
Agentic orchestration	Coordinates the reasoning steps into a result	Synthesis agent (the LLM answer-and-cite chain)

Why this matters for defensibility

Choosing a problem that matched an already-operating architecture meant the build risk was in execution, not in learning an unfamiliar stack under a deadline. The design decisions here are the same ones already validated in production, applied to a new corpus.

7. Dataset

The evaluation corpus is engineered, not just collected, so that every judging metric has a source that exercises it. It centres on one asset class — centrifugal pumps and their drive motors — so relationship chains run deep rather than shallow.

Layer	Contents and purpose
Synthetic corpus	11 generated PDFs built around a deliberately planted five-document near-miss failure chain, plus realistic distractor documents that give keyword search plausible-but-wrong matches.
Real OEM manuals	3 genuine centrifugal-pump manuals (operation, maintenance, spec tables) — real formatting, real equipment tags, real messiness.
Scanned documents	3 degraded image versions of synthetic documents, to exercise the OCR path end-to-end.
Evaluation set	FailureSensorIQ (IBM Research, NeurIPS 2025, CC-BY-4.0): 8,296 industrial QA pairs over 10 asset types, used as an external reference benchmark layer.
Ground truth	Hand-labelled entity set across three document types, enabling a real Macro-F1 rather than an estimated one, plus eight benchmark questions tagged by retrieval type.

The planted multi-hop chain

Five documents describe a single pump failure, each holding one link of the causal chain, and no two sharing the keywords that would let search connect them:

- **Inspection report (day −21):** elevated drive-end bearing vibration flagged, status 'monitor'.
- **Maintenance log (day −14):** lubrication topped up, bearing replacement deferred due to a missing spare — against procedure.
- **Vibration trend log (day −5):** vibration crosses the alarm threshold; no work order raised.
- **Incident report (day 0):** catastrophic bearing seizure, unplanned downtime.
- **Procedure and regulation:** specify what should have happened after the first warning.

> **The star demo question**
>
> "Was there any early warning before the Pump P-204 failure, and what should have been done according to procedure?" Pure keyword or vector search returns only the incident report. Graph traversal recovers the full warning chain — which is the entire point of the hybrid architecture, demonstrated live.

8. Evaluation and Results

All figures below are final, reproducible, and measured against hand-labelled ground truth. They are reported as measured, with limitations stated plainly rather than smoothed over.

Entity extraction accuracy

Metric	Value	Note
Macro-F1 (9 entity types)	0.8411	Excludes DATE — see below
Recall across gold entities	Strong	Only 3 misses across 39 gold entities
Precision	Moderate	Main loss is duplicate surface forms of the same entity

DATE is excluded from the Macro-F1 because the ground-truth set contains zero labelled date instances in these documents, which makes recall for that type mathematically undefined; including it as a defaulted zero would penalise the system for a category never under test. The dominant precision loss is benign: the extractor captures the same real entity under two surface forms (for example both the bare tag and the fully-qualified name), rather than inventing facts that are not present.

Retrieval quality (hybrid versus keyword baseline)

Metric	Value	Type
Star demo question (Q01) recall	0.60	Multi-hop, graph-only
Overall hybrid recall (8 questions)	0.78	Mixed types
Control questions (keyword-answerable)	1.00	Unchanged, as expected

The control questions confirm the hybrid system loses nothing on simple lookups, while the multi-hop questions are where graph traversal adds recall that keyword search structurally cannot reach. Retrieval is fully deterministic: repeated runs on the same inputs produce byte-for-byte identical rankings.

9. Known Limitations

These are documented deliberately. Understanding a system's failure modes precisely is more valuable than hiding them, and each has a traced root cause.

- **Non-determinism, found and fixed.** Early benchmark numbers varied across runs of identical code. The cause was unstable tie-break ordering in graph ranking, sensitive to per-process hash randomisation. It was fixed by adding a deterministic final tie-break key, and verified by repeated identical runs. Every metric reported here is post-fix.
- **Hop-tier tie-flooding (Q02).** For one question, many documents share an identical graph proximity score simply for mentioning the query's main entity, which can crowd out a more specific but structurally-further answer. Two targeted fixes were attempted and precisely diagnosed; the residual gap is understood and documented rather than papered over.
- **Supporting-document cutoff (Q01).** The star question answers correctly with the core chain, but one supporting document occasionally ranks just outside the retrieval cutoff — the same tie-flooding pattern. The answer's narrative remains correct; one corroborating detail can be omitted.

10. Scalability and Roadmap

The prototype runs entirely local, with only the language-model calls hitting a network. Each stage is modular and has a clear production upgrade path: the in-memory graph swaps for a graph database, the local vector store for a managed vector service, without touching the rest of the pipeline. Ingestion is incremental — new documents merge into the existing graph and vector store without a full rebuild — which satisfies the 'continuously updated' requirement.

Three further components are specified for the next phase: a maintenance and root-cause agent that turns failure history into predictive recommendations; a regulatory compliance mapper that checks procedures against standards and flags gaps; and a lessons-learned engine that surfaces recurring patterns across incident records. Each reuses the same ingestion and graph foundation already built.

11. Conclusion

This submission is a working, benchmarked, honestly-documented Hybrid GraphRAG system that answers the core of Problem #8: it turns scattered industrial documents into a single queryable intelligence layer, and it proves the difference from keyword search live, on a multi-hop question that only relationship reasoning can answer. The metrics are real and reproducible, the limitations are understood and stated, and the architecture is one already validated in production — re-pointed at a new domain rather than assembled from scratch under a deadline.

Reproducibility

The full pipeline runs from a public repository with a fixed seed. The corpus, benchmark questions, ground-truth labels, and evaluation harness are all included, so any reviewer can clone the repository and reproduce the reported numbers.